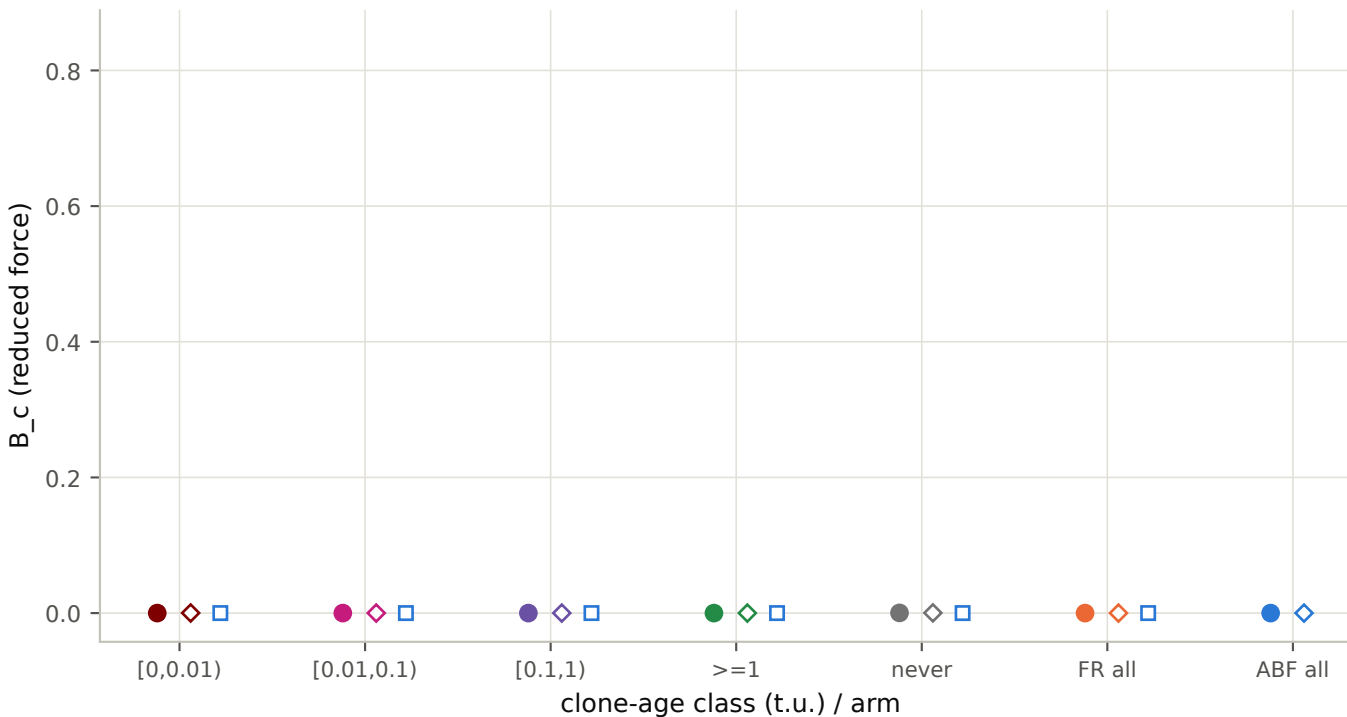
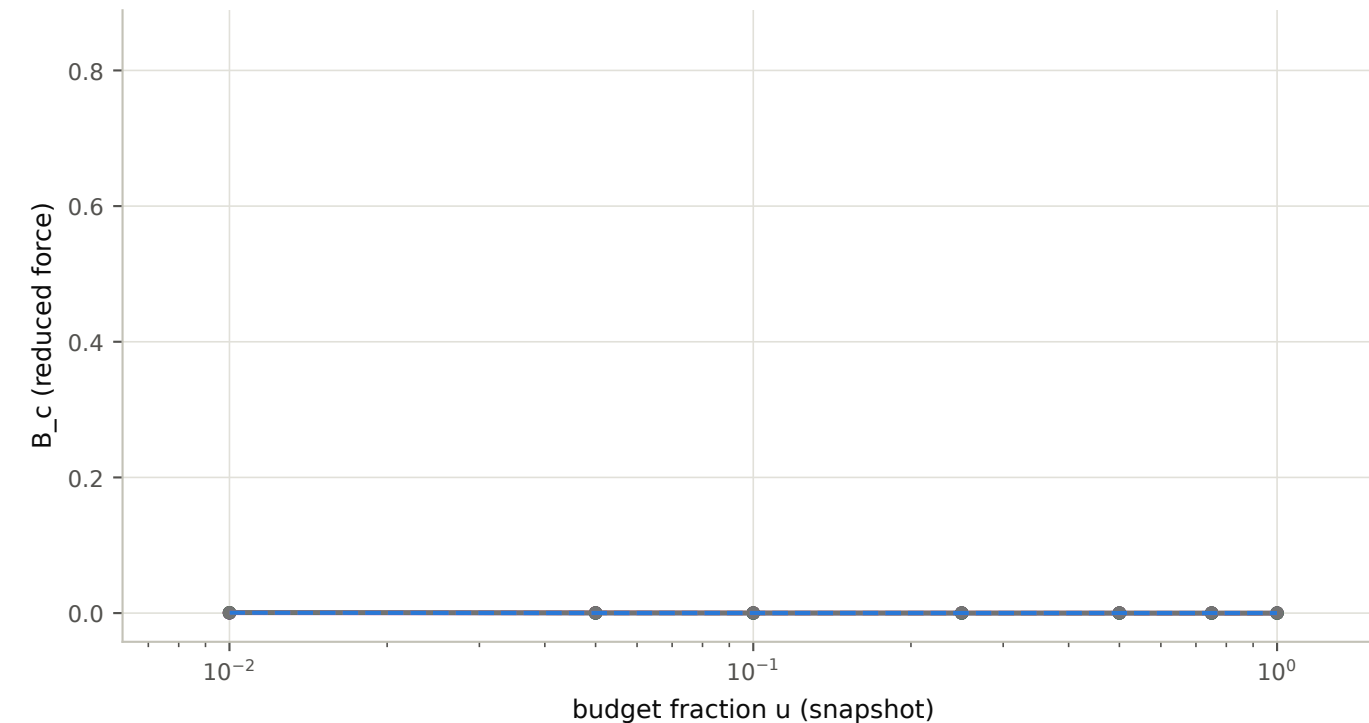


D2(ii) class-wise conditional mean-force error vs the bin-averaged F^* : I : $\alpha = 0$ (purely energetic barrier in x ; deterministic force), $N = 512$
 $B_c^2 = \sum_j n_{cj} [(m_{cj} - \langle F^* \rangle_j)^2 - V_{cj}] / \sum_j n_{cj}$ over eval bins (V_{cj} : seed-cluster variance of the pooled bin mean); CIs: 10000 seed-bootstrap resamples (ABF+FR $n = 32$, ABF $n = 32$; B_c : basic interval of the debiased B_c^2). The ABF arm is the reference level. y ranges shared by every cell of this experiment at this N .

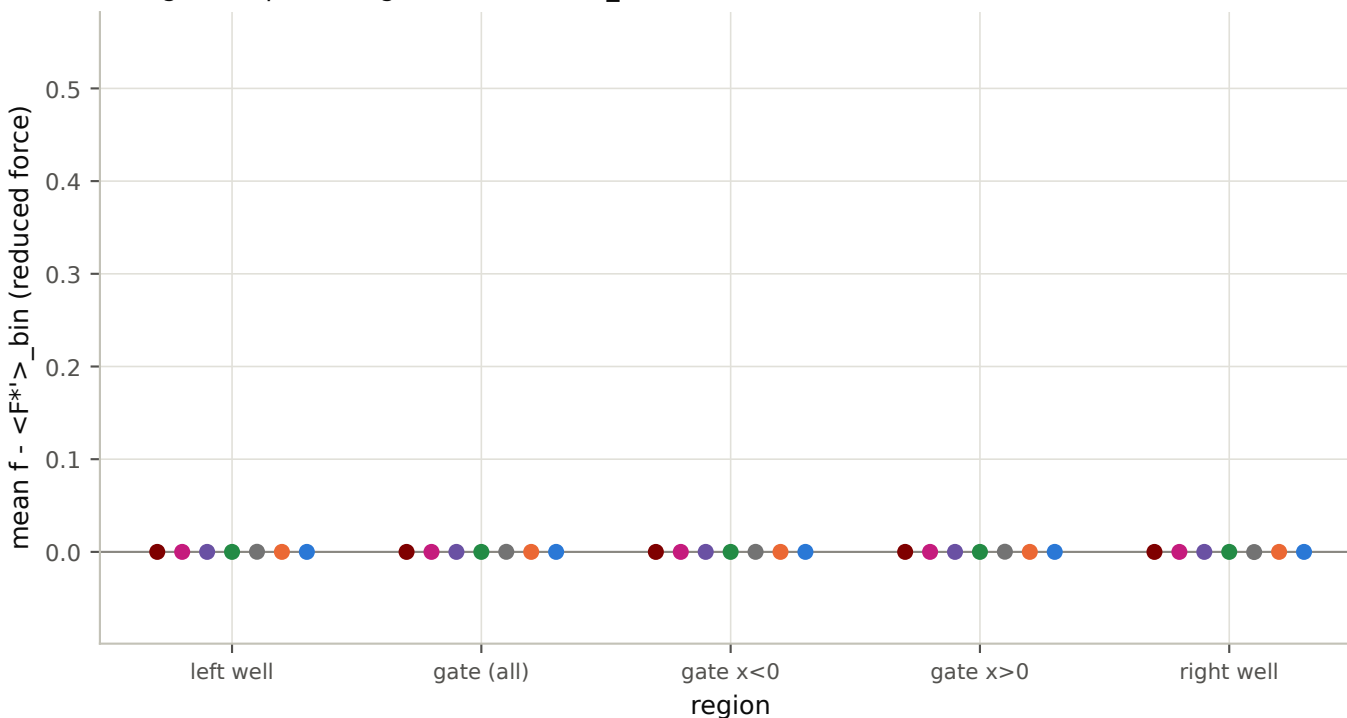
(a) debiased RMS mean-force bias B_c at the end



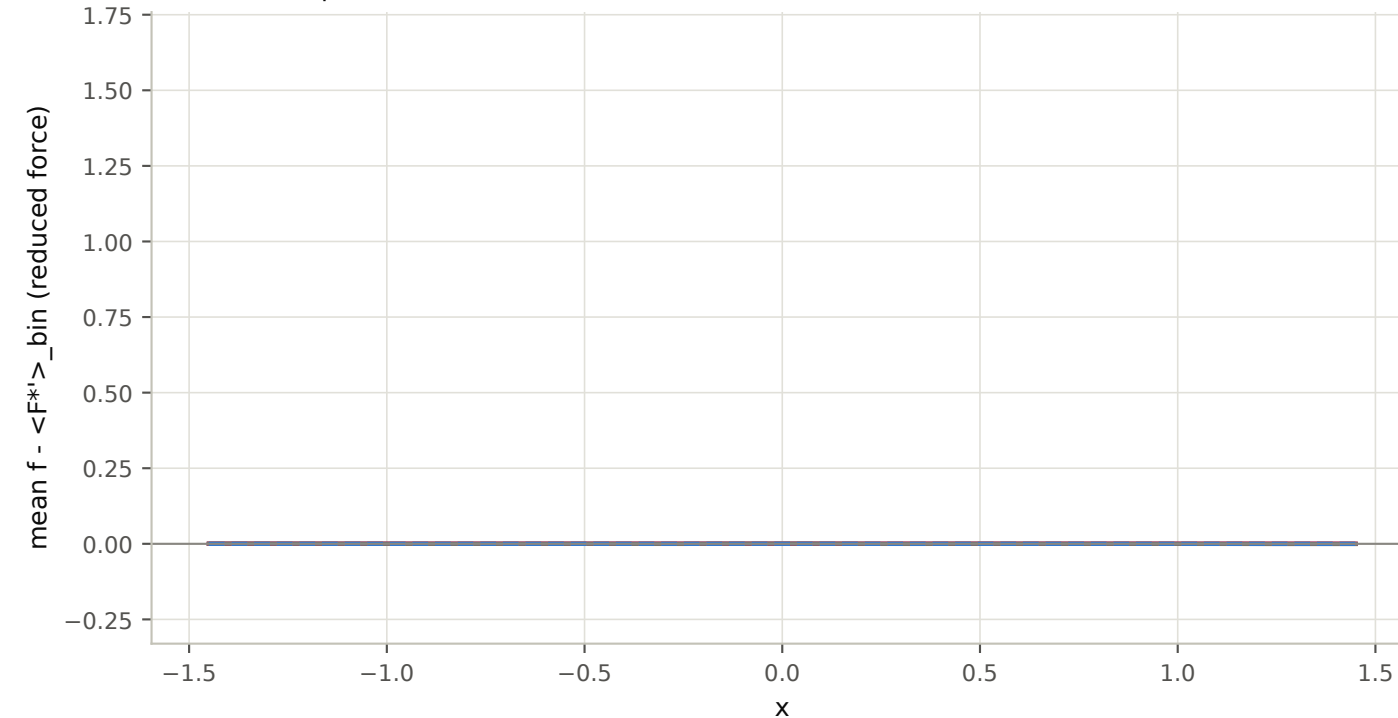
(b) B_c of the cumulative deposits at each snapshot



(c) signed deposit-weighted deviation D_{cR} at the end (eval window)



(d) deviation profile at the end (0.1-wide bins)



— clone age [0,0.01) t.u. — clone age [0.1,1) t.u. — never cloned — ABF (all deposits = never cloned) □ (a) EXPLORATORY: ABF's B at the class's positions (matched, paired)
— clone age [0.01,0.1) t.u. — clone age ≥ 1 t.u. — ABF+FR, all deposits ◇ raw RMS (not debiased) 95 % seed-bootstrap CI